

Customer Churn Prediction

Professionalized machine learning repository report

Problem

Predict whether a bank customer is likely to leave using account, demographic, and activity signals.

Workflow

Raw dataset -> preprocessing -> feature engineering -> model comparison -> evaluation outputs -> saved model -> Streamlit app.

Models Compared

Model	Purpose
Logistic Regression	Baseline comparison candidate
Random Forest	Baseline comparison candidate
Gradient Boosting	Baseline comparison candidate

Dataset Columns

Column
CreditScore
Age
Tenure
Balance
NumOfProducts
IsActiveMember
Exited

Repository Improvements

- Organized folders for data, notebooks, source code, models, outputs, app, docs, images, and reports.
- Moved reusable logic into modular Python files.
- Added config.yaml for reproducible parameters.
- Added logging-oriented training scripts and output artifacts.
- Added documentation for business context, architecture, data dictionary, deployment, and usage.

Key Notes

- High churn risk is often linked with lower activity and less stable account relationships.
- The Streamlit app returns churn probability and downloadable predictions.

Next Steps

- Run training after installing dependencies.
- Capture Streamlit screenshots for the images folder.
- Add SHAP or feature contribution explanations if presentation time allows.